

Ag (Ag I). Strongest classified transitions below 2 μm (NIST): 29 levels, 62 lines.

[illegible]

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ (J er J') $ (eao)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4d^{10}5s\ ^2S_{1/2} - 4d^95s^2\ ^2D_{3/2}$	288.0663	287.9818	1040.7063	-	1.130e-05	0.003979	0	two-photon line at 576 nm
$4d^{10}5s\ ^2S_{1/2} - 4d^{10}5p\ ^2P_{3/2}$	328.1625	328.0680	913.548766(20)	3.206	1.473e+08	23.44	100	D2 resonance line
$4d^{10}5s\ ^2S_{1/2} - 4d^95s^2\ ^2D_{5/2}$	330.6628	330.5676	906.641024	-	5.220e+00	9.764e-07	85.1	ultra-narrow E2 clock transition
$4d^{10}5s\ ^2S_{1/2} - 4d^{10}5p\ ^2P_{1/2}$	338.3859	338.2888	885.9484	2.272	1.350e+08	21.48	100	D1 resonance line
$4d^95s^2\ ^2D_{5/2} - 4d^{10}6p\ ^2P_{3/2}$	547.6898	547.5376	547.3764	0.233*	1.668e+05	2.04	1.3	547.7 nm line used for spectroscopy of the metastable $2D5_{3/2}$

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

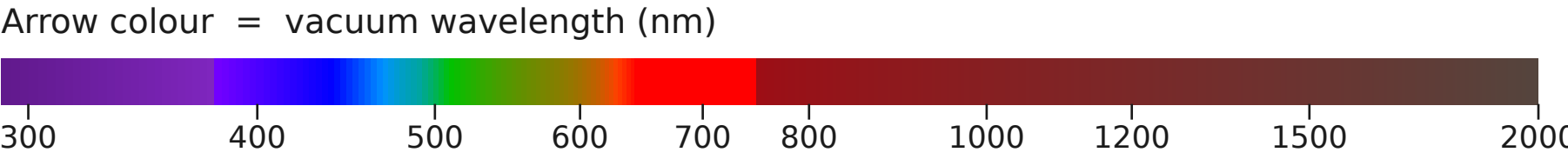
$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| e r \| J' \rangle|$ in e.

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: NIST energy and measured lifetime (where one exists).



Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$4d^{10}5s\ ^2S_{1/2}$	-1712.512111(18)	-	F=0: +1284.38 F=1: -428.13
$4d^{10}5p\ ^2P_{1/2}^o$	-175.4(17)	-	F=0: +131.55 F=1: -43.85
$4d^95s^2\ ^2D_{5/2}$	-126.28180(10)	-	F=2: +220.99 F=3: -157.85
$4d^{10}5p\ ^2P_{3/2}^o$	-31.7(5)	-	F=1: +39.62 F=2: -23.77
$4d^{10}6p\ ^2P_{1/2}^o$	-38.7(10)	-	F=0: +29.02 F=1: -9.68
$4d^{10}6p\ ^2P_{3/2}^o$	-9.05(25)	-	F=1: +11.31 F=2: -6.79
$4d^{10}7p\ ^2P_{3/2}^o$	-4.50(20)	-	F=1: +5.62 F=2: -3.38

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$4d^{10}5s\ ^2S_{1/2} - 4d^{10}6p\ ^2P_{3/2}$	206.182	107-109	410.0(6)
$4d^{10}5s\ ^2S_{1/2} - 4d^{10}6s\ ^2S_{1/2}$	234.984	107-109	368.7(0)
$4d^{10}5s\ ^2S_{1/2} - 4d^{10}5p\ ^2P_{3/2}$	328.163	107-109	473.2(7)
$4d^{10}5s\ ^2S_{1/2} - 4d^95s^2\ ^2D_{5/2}$	330.663	109-107	564.1(4)
$4d^{10}5s\ ^2S_{1/2} - 4d^{10}5p\ ^2P_{1/2}$	338.386	107-109	473(4)
$4d^95s^2\ ^2D_{3/2} - 4d^{10}6p\ ^2P_{3/2}$	547.690	109-107	-978.1(5)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$